

DRIVEGO CURRENT STATE MASTER AUDIT

ARCHITECTURE • UI/UX • INTEGRATION RECONCILIATION • 2 SEPTEMBER 2026

1. GIT BASELINE & VERIFICATION SIGN-OFF

Active Branch	main	Working Tree	Clean (No code changes)
Current Local HEAD	3b8fdc338b94396255a839be307bfa0074	Sync Status	HEAD == origin/main (Synchronized)
Phase 29.11 Commits	3b8fdc3 (Vendor Location Engine)	Phase 29.10 Commits	bfc814b (Inspection & Handover)
Audit Directive	NO-CODE-CHANGE / Evidence Only	Target Platform	Flutter 3.x / NestJS / PostgreSQL / Redis

2. REQUIREMENTS SOURCE OF TRUTH (SIX ARCHITECTURE DOCS)

Document	Scope & Core Contract	Reconciliation Status
01_PRD.docx	Product Contract, 7 Handover Modes, Availability & Booking Rules	GREEN (Mapped)
02_TRD.docx	Technical Contract, Thin Controllers, Central Eligibility, Pricing Math	YELLOW (Partially Met)
03_App_Flow.docx	Master Journey: Search -> Eligibility -> Quote -> Snapshot -> Handover	YELLOW (String Gap)
04_UIUX_Brief.docx	DriveGo Design System (DDS) Mobile & Admin Viewport Specs	GREEN (DDS Compliant)
05_Backend_Schema.docx	10 Logical Entities: Location, Hours, Exceptions, Policy, Quote, Snapshot	BLUE (Schema Overloaded)
06_Impl_Plan.docx	8-Phase Controlled Execution Sequence, Testing Gates & Stop Conditions	GREEN (Documented)

3. EXECUTIVE ARCHITECTURE FINDINGS

- Customer App is fully modernized with DDS tokens across 23 routes; search & booking are operational on live backend.
- Phase 29.11 Vendor Location UI is visually complete (17 native AVD screenshots verified, 33/33 tests passing).
- **CRITICAL ARCHITECTURAL GAP: Delivery policies and one-way relocation matrices are stored in transient Redis keys only.**
- **CONTRACT MISMATCH: Vendor App calls /delivery-policy (404) with PATCH (405) against backend /policy (PUT) with silent catches.**
- CUSTOMER INTEGRATION GAP: Customer checkout degrades locations to plain strings and bypasses live /locations/quote math.
- ADMIN CONTROL TOWER: 22 feature pages exist, but layout lacks domain grouping and dedicated Location Review Governance.

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SECTION 7: LOCATION & FULFILLMENT RECONCILIATION • 2 SEPTEMBER 2026

7. REQUIREMENTS VS. ACTUAL IMPLEMENTATION SCORECARD

Entity / Requirement	Status	Implementation Reality & Gap Analysis
VendorLocation	YELLOW	PickupHub model reused; metadata serialized as JSON in operatingHours column.
LocationHours	YELLOW	Stored in JSON blob inside PickupHub.operatingHours; no separate SQL joinable table.
LocationException	RED	Missing completely; no holiday or temporary closure date model in schema.
LocationCapability	YELLOW	allowsPickup/Return/Delivery stored in JSON blob; cannot be indexed for spatial search.
ServiceArea	GREEN	serviceRadiusKm on PickupHub and maxDeliveryRadiusKm enforced in quote calculations.
VehicleLocationAssignment	GREEN	Car.pickupHubId foreign key relation with batch assignment API verified.
FulfillmentRule	BLUE	Stored in Redis cache key (vendor:delivery-policy:<id>) only; no PostgreSQL table.
FulfillmentQuote	YELLOW	Backend math complete in /locations/quote; bypassed by Customer App Checkout.
BookingFulfillmentSnapshot	YELLOW	Flat fields on Booking table; missing oneWayFee snapshot column and hub FKs.
LocationAuditEvent	GREY	Generic AuditLog exists; vendor location CRUD updates currently bypass audit log.

8. CUSTOMER FULFILLMENT INTEGRATION TRACE (HOME -> HANDOVER)

- Customer Home: User selects City and Date Range (Real API).
- Location Selector: User selects location string from hardcoded list (_cityPopularHubs) -> DATA DEGRADES TO PLAIN STRING.**
- Search & Results: Filters cars by city & dates; passes plain string pickup/drop locations.
- Vehicle Details: Displays vehicle specifications and mileage package pricing.
- Checkout Flow (Step 1-5): Computes rental fare; DOES NOT CALL /locations/quote (Zero delivery or one-way fee added).**
- Booking Confirmation: Booking created in DB with flat string pickup/drop location fields.
- Vendor Handover: Vendor sees plain pickup location string; conducts 4-photo inspection & verifies customer OTP.

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SECTION 17: FINAL CTO STRATEGIC DECISION • 2 SEPTEMBER 2026

17. AUTHORITATIVE CTO ANSWERS TO THE 10 KEY QUESTIONS

1. Genuinely Production-Ready	Customer Auth, Profile, KYC uploads, Vehicle Search by City/Dates, Mileage Packages, Core Bookings, Vendor Registration, Fleet Management, Fast Add, CSV Bulk Upload, 4-Photo Pre/Post Trip Inspections, Odometer Monotonicity, and OTP Handover Protocols.
2. UI-Complete Only	Vendor Location Operations Wizard (Phase 29.11) and Admin Operational Map. UI is pristine and fully responsive, but backed by overloaded JSON strings and static fallback data.
3. Mock / Fallback Elements	Public Location Catalog (in-memory static array), Customer Popular Hubs (<code>_cityPopularHubs</code>), Vendor Initial State (<code>_getStaticMockLocations</code>), Backend Summary Fallbacks (<code> 8, 4</code>), and Wallet Mock Payment Signature.
4. Architecturally Incomplete	Customer-to-Vendor Fulfillment Integration (Checkout does not request <code>/locations/quote</code>), Delivery Policy Persistence (Redis-only storage), and Admin Location Governance.
5. Top 3 Highest Risks	(1) Redis restart will wipe out vendor delivery configurations; (2) Silent catches in Vendor App conceal 404/405 API failures; (3) Customers are not charged delivery/one-way surcharges.
6. Top 3 Highest Value UI Fixes	(1) Wire dynamic delivery quote calculation into Customer Checkout; (2) Fix URL/method mismatches in Vendor App; (3) Upgrade Admin Operational Map into a Location Governance Control Tower.
7. Recommended Next Phase	PHASE 29.12: Location & Fulfillment End-to-End Integration & Persistence Hardening (Persist rules to PostgreSQL, align REST contracts, wire Customer Checkout quote, and capture snapshot columns).
8. What NOT to Touch Yet	Do NOT modify Fleet Management (29.8/29.9), Handover/Inspection (29.10), or Customer Checkout core state machine (29.4/29.5).
9. Admin Control Tower Decision	Defer Admin Control Tower modernization until immediately after Phase 29.12 to prevent building governance UI on top of transient Redis contracts.
10. Location Completion First?	YES. Establishing verified, normalized PostgreSQL persistence for locations and delivery quotes is a mandatory prerequisite for Admin Control Tower governance.

AUDIT CONCLUSION: ARCHITECTURE BASELINE VERIFIED • PROCEED WITH PHASE 29.12